

AI INTERNSHIP - WEEK 2

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1. **Features vs. labels** — features (inputs, X) are what you know; the label (y) is what you predict.

- features(x): or inputs, they are independent and available to be used for training the models
- label(y): or outputs, they are the desired value/goal of the model

2. **Train / validation / test split** — why we split three ways and never let the model see test data during training (you don't give students the exam while they study).

training: fitting the model and teaching it based upon provided data to check patterns and provide the most accurate prediction

validation: checks the model for overfitting during the training

test: the unseen data on which the model predicts how the model will perform in real life

split: splitting the data into different categories for different acts to be performed upon it (80% training, 20% test)

3. • **Overfitting vs. underfitting** — memorizing noise vs. too simple to capture the pattern.

overfitting: giving the model all the data for training, so it memorizes it and performs well in training but becomes bad when faced with unseen data

underfitting: the model performs poorly in both training and testing phase, it's over simplified so it can't grasp any patterns

4. • **Bias–variance trade-off** — one paragraph, plain language.

bias means the data is (underfitted) because it's way too simple and pays no real attention to the details, while variance means the data is overfitted because the model is too complex or was training for long so it pays attention to every single detail

the trade off means to balance them out, no high bias neither high variance, to fix the fitting issues